DHCP Client Simulation

• Real-World Relevance: Understand how devices automatically get IP addresses on a network, a process every student uses daily.

• Core Concepts: DHCP DORA process (Discover, Offer, Request, Acknowledge), UDP broadcasting.

• 10-Week Plan:
o Weeks 1-2: Learn to create raw sockets and send/receive broadcast packets.
o Weeks 3-5: Craft a DHCP Discover packet and broadcast it.
o Weeks 6-7: Parse the DHCP Offer response from a server (e.g., a home router).
o Weeks 8-10: Craft and send a DHCP Request packet and process the final
Acknowledgment to obtain a valid IP lease.

• Tools: Python, Scapy (highly recommended for crafting packets), Wireshark for analysis.